NEET-SS / FET / INI-SS HIGH-YIELD NOTES

Paediatric Ortho

1. DDH
2. Perthes and SCFE
3. clubfoot and foot deformities
4. paediatric fractures
5. cerebral palsy and neuromuscular
6. skeletal dysplasias
7. bone and joint infection in children

DDH

Who gets it

- Risk factors: **breech** (strongest), female, first-born, family history, oligohydramnios, tight swaddling with hips extended. **Left hip** in about 60%.
- Associated "packaging" conditions: torticollis, metatarsus adductus, congenital knee dislocation.

Examination by age

- **Newborn: Barlow** (adduct + push posteriorly: dislocatable hip) and **Ortolani** (abduct + lift trochanter: dislocated hip reduces with a clunk).
- **After about 3 months:** these become negative. **Limited abduction** is the most reliable sign; **Galeazzi** (unilateral, knee lower); Klisic line.
- **Walking child:** Trendelenburg limp (unilateral), waddling gait and hyperlordosis (bilateral).

Imaging

- **Ultrasound** until the femoral head ossifies (about 4-6 months). **Graf alpha angle** of 60 degrees or more is normal (type I). Type IIa (50-59 degrees, under 3 months) is physiological immaturity: re-scan. Lower angles and types III/IV need treatment. Femoral head coverage under 50% is abnormal.
- **Radiograph** after 4-6 months: Hilgenreiner and Perkin lines (head should lie in the inferomedial quadrant), broken **Shenton line**, raised **acetabular index**, delayed ossific nucleus. In older children and adults, a **centre-edge angle** under 20 degrees means dysplasia.

Treatment by age

Age	Treatment
0-6 months	**Pavlik harness**: hips flexed 90-100 degrees, abduction not forced. Confirm reduction on ultrasound; stop if not reduced in 3-4 weeks
6-18 months (or Pavlik failure)	**Closed reduction** under anaesthesia with arthrogram, adductor tenotomy, spica in the "human position" (about 100 degrees flexion, abduction within the Ramsey safe zone)
over 18 months	**Open reduction**, often with femoral shortening and pelvic osteotomy
beyond about 8 years unilateral (6 bilateral)	Usually leave a high dislocation unreduced

- **Pavlik complications:** femoral nerve palsy (excess flexion), inferior dislocation, AVN (forced abduction), "Pavlik disease" (posterior wall erosion from persisting with a non-reduced hip).
- **Pelvic osteotomies: Salter** (redirection, 18 months-6 years, needs a concentric reduction); **Pemberton** (reshaping, capacious acetabulum, open triradiate cartilage); **Dega** (posterior deficiency, neuromuscular hips); after triradiate closure, **Ganz periacetabular osteotomy**.

Blocks to reduction

- **Extra-articular:** iliopsoas (hourglass capsule), adductor longus.
- **Intra-articular:** pulvinar, hypertrophied ligamentum teres, transverse acetabular ligament, inverted labrum (limbus).
- The **medial (Ludloff) approach** in infants endangers the medial femoral circumflex artery.

Exam pearl: **AVN of the femoral head** is the most important complication of treatment, not of the disease itself (Kalamchi-MacEwen classification).

Perthes and SCFE

Perthes disease (idiopathic AVN of the capital epiphysis)

- Age **4-8**, boys 4-5:1, bilateral in about 10-15% (asynchronous), delayed bone age. Symmetrical bilateral changes suggest **multiple epiphyseal dysplasia** or hypothyroidism instead.
- **Waldenstrom stages:** initial (dense epiphysis, **crescent sign**), fragmentation, reossification, healed.

Classification	What it tells you
Herring lateral pillar (fragmentation stage)	A: pillar height normal. B: over 50% preserved. B/C border: about 50%, or a narrow or poorly ossified pillar. C: under 50%. Best predictor of outcome, together with age
Catterall I-IV	Extent of head involvement; "head at risk" signs: **Gage sign**, lateral calcification, lateral subluxation, horizontal physis, metaphyseal cysts
Salter-Thompson	Extent of the subchondral fracture: A under 50%, B over 50%
Stulberg (at maturity)	I-II spherical and congruent (good); III-IV aspherical congruence; **V aspherical incongruence** (worst)

- **Prognosis: age at onset** is the most important factor (under 6 years does well), then lateral pillar class. Girls fare worse.
- **Treatment = containment.** Under 6 years: symptomatic care and range of motion. **Over 8 years with lateral pillar B or B/C:** surgical containment (proximal femoral varus osteotomy or Salter innominate osteotomy) gives better heads (Herring's multicentre study).

- **Hinge abduction:** valgus-extension osteotomy or shelf acetabuloplasty (salvage).

Slipped capital femoral epiphysis (SCFE)

- Obese adolescents: boys 13-15, girls 11-13. Bilateral in 20-40%.
- Suspect an **endocrine cause** (hypothyroidism, growth hormone therapy, renal osteodystrophy) if under 10, over 16, or thin.
- The metaphysis moves anterior and externally rotated; the epiphysis stays posterior and medial in the acetabulum.
- **Signs:** obligatory external rotation on flexion (**Drehmann sign**); knee or thigh pain (obturator nerve referral). Never miss it in a child with knee pain.
- **Loder stability** (the key prognostic split): **stable** = can walk, AVN risk near 0; **unstable** = cannot walk even with crutches, AVN risk up to about 50%.
- **Imaging: Klein line** does not cross the epiphysis (Trethowan sign); Steel metaphyseal blanch sign. **Southwick slip angle** on the lateral view: mild under 30 degrees, moderate 30-50, severe over 50. The frog-leg lateral is the most sensitive view, but avoid it in an unstable slip.
- **Treatment: in situ fixation with one cannulated screw**, centred in the epiphysis, perpendicular to the physis, with no joint penetration. Do not reduce a stable slip.
- Prophylactic fixation of the other hip: endocrine cause, age under 10, open triradiate cartilage, poor follow-up.
- Severe or unstable slips: **modified Dunn** realignment through a surgical dislocation, in experienced hands.
- **Complications:** AVN (unstable slip, forceful reduction), **chondrolysis** (screw penetration), cam-type FAI.

clubfoot and foot deformities

Congenital talipes equinovarus (clubfoot)

- Deformity **CAVE**: **C**avus (plantarflexed first ray), forefoot **A**dductus, hindfoot **V**arus, **E**quinus.
- The talar neck is deviated medially and plantarward; the navicular is displaced medially. Twice as common in boys; half are bilateral.
- **Pirani score** (0-6: 3 midfoot + 3 hindfoot) and Dimeglio grade assess severity and track correction.

Ponseti method

1. Correct the **cavus first** by **supinating the forefoot** (lift the first metatarsal) to line it up with the hindfoot.
2. **Abduct** the supinated foot around the **head of the talus** as fulcrum. Never press on the calcaneocuboid joint (**Kite's error**), and never pronate.
3. Weekly long-leg casts, usually 5-6.
4. **Percutaneous Achilles tenotomy** (needed in about 80-90%) once abduction of about 60-70 degrees is reached but ankle dorsiflexion stays under about 10-15 degrees; then 3 weeks in a cast.
5. **Foot abduction brace**: 70 degrees external rotation on the clubfoot side (about 40 on a normal side), 10-15 degrees dorsiflexion; 23 hours a day for 3 months, then nights and naps until 4-5 years.

- **Relapse** is mostly due to stopping the brace. Re-cast. Dynamic supination in a child over about 3 years: **tibialis anterior transfer to the lateral cuneiform**.
- **Complex (atypical) clubfoot:** short, fat foot, deep plantar crease, short hyperextended big toe. Use the modified Ponseti

technique.

Other foot deformities

- **Congenital vertical talus (rocker-bottom):** irreducible dorsal dislocation of the navicular on the talus. **Forced plantarflexion lateral view:** the talar axis still passes plantar to the first metatarsal and cuneiform (**oblique talus** reduces). Treat with **reverse Ponseti (Dobbs)** casting, then limited open pinning of the talonavicular joint and Achilles tenotomy. Associated with arthrogryposis, myelomeningocele and chromosomal syndromes.
- **Metatarsus adductus:** a kidney-shaped foot with a flexible heel; the **heel bisector** (Bleck) grades it. Most resolve; stretch or cast the rigid ones.
- **Tarsal coalition** (peroneal spastic flatfoot, recurrent ankle sprains):
  - **Calcaneonavicular:** age 8-12, **"anteater" sign** on the oblique view. Excise and interpose EDB or fat.
  - **Talocalcaneal (middle facet):** age 12-16, **C-sign** on the lateral view; CT is the test of choice. Resect (with fat interposition) if the coalition is under about 50% of the subtalar joint surface and there is no arthritis; otherwise fuse.
- **Flexible flatfoot** is physiological if the arch appears on tiptoe and the hindfoot swings into varus. Treat only if painful.
- **Cavovarus foot in a child:** look for a neurological cause (Charcot-Marie-Tooth, spinal dysraphism). MRI of the spine if unilateral or progressive.

paediatric fractures

1. Why children's fractures are different

- **Periosteum** is thick, strong and osteogenic: it limits displacement, acts as a hinge for reduction (keep the intact periosteal side as the hinge) and produces fast, abundant callus.
- **Bone is more porous and less brittle:** it fails in compression (**torus/buckle**), on the tension side only (**greenstick**), or bends without fracture (**plastic deformation**, most often ulna and fibula; can block reduction of the partner bone).
- **Remodelling** is greatest when the child is young, the fracture is **near a fast-growing physis**, and the deformity is **in the plane of motion of the adjacent joint**. **Rotational deformity does not remodel.** Intra-articular step-off does not remodel.
- **Overgrowth:** femoral shaft fractures in children aged about 2-10 overgrow by roughly 1-2 cm, which is why 1-2 cm of shortening (overlap) is accepted in a spica.
- **Ligaments are stronger than the physis:** an apparent "sprain" in a child is often a physeal injury (e.g. Salter-Harris I of the distal fibula).
- **Fastest-growing physes:** proximal humerus (about 80% of humeral growth) and distal radius (about 75-80% of radial growth) in the upper limb; **distal femur (about 70% of femoral length)** and proximal tibia (about 55-60% of tibial length) around the knee. Roughly 1 cm a year at the distal femur and 0.6 cm at the proximal tibia.

2. The physis and Salter-Harris

- **Zones (epiphysis to metaphysis):** reserve (resting), proliferative, **hypertrophic** (maturation, degeneration, **provisional calcification**). Fractures usually propagate through the hypertrophic zone, the weakest region; the germinal (reserve and proliferative) cells stay with the epiphysis, which is why most physeal fractures grow normally.
- **Salter-Harris classification:**

Type	Fracture line	Notes
I	Through the physis only	Infants and young children; radiographs may be normal (look for widening); distal fibula, proximal humerus in neonates
II	Physis + metaphyseal fragment (**Thurston-Holland**)	**Commonest (about 75%)**; the periosteum is intact on the fragment side
III	Physis + epiphysis (intra-articular)	Needs anatomical reduction (e.g. juvenile Tillaux)
IV	Metaphysis + physis + epiphysis	Intra-articular and crosses the germinal layer; **ORIF**; high risk of a bar (e.g. medial malleolus, lateral condyle humerus)
V	Crush of the physis	Usually diagnosed retrospectively when growth arrest appears; worst prognosis
VI (Rang)	Perichondral ring injury	Peripheral bar, angular deformity (lawnmower injuries)

- **Peterson type I** (transverse metaphyseal fracture extending into the physis, distal radius) is worth knowing for FET-level questions.

- **Highest growth-disturbance risk: distal femur** (even SH I and II, because of the undulating physis), proximal tibia, distal tibia SH III/IV.
- **Physeal bar:** diagnose with **MRI (or CT) mapping**. Resect with interposition (fat, cement) if it involves **under 50%** of the physis and the child has **at least 2 years (about 2 cm) of growth left**. Otherwise complete the arrest (epiphysiodesis), deal with the other bone, and correct the angulation and length.
- **Park-Harris growth arrest lines** that converge on the physis indicate a bar; parallel lines mean symmetrical growth.

3. Elbow (the high-yield region)

Ossification centres (CRITOE)

- **C**apitellum about 1 year, **R**adial head 3-5, **I**nternal (medial) epicondyle 5-7, **T**rochlea 7-9, **O**lecranon 9-11, **E**xternal (lateral) epicondyle 11-13 (girls earlier).
- The **medial epicondyle always appears before the trochlea**: an ossified fragment in the joint before the trochlea ossifies is an incarcerated medial epicondyle.
- **Anterior humeral line** passes through the middle third of the capitellum; the **radiocapitellar line** passes through the capitellum in every view (if not, look for Monteggia).
- **Posterior fat pad sign** = effusion = occult fracture (supracondylar, lateral condyle, radial neck).

Supracondylar humerus fracture

- Commonest elbow fracture in children (peak 5-7 years), mechanism fall on the outstretched hand. **Extension type about 97%**, flexion type about 3%.
- **Gartland classification (with Wilkins modification):**

Type	Description	Treatment
I	Undisplaced; posterior fat pad only	Above-elbow cast 3 weeks
IIA	Displaced, posterior cortex intact (hinge), no rotation	Closed reduction; pin if anterior humeral line misses the capitellum
IIB	Hinged but rotated or translated	Closed reduction and pinning
III	Completely displaced	**Closed reduction and percutaneous pinning**
IV (Leitch)	Multidirectionally unstable (periosteal hinge lost)	Pinning; often diagnosed under the image intensifier

- **Nerves: anterior interosseous nerve is the commonest** (extension type), then radial, median and ulnar.
  - **Posterolateral displacement** of the distal fragment (proximal spike goes anteromedial): **median nerve and brachial artery**.
  - **Posteromedial displacement** (spike goes anterolateral): **radial nerve**.
  - **Flexion type: ulnar nerve**.
  - **Iatrogenic:** ulnar nerve with a medial pin, especially when the elbow is hyperflexed (the nerve subluxes anteriorly).
  - Most nerve palsies are neurapraxias that recover in 2-3 months.
- **Pinning: 2 or 3 divergent lateral pins** separated at the fracture site are stable for most fractures with no ulnar nerve risk. Crossed pins are biomechanically stiffer; if a medial pin is needed, make a small incision and extend the elbow before placing it.
- **Vascular status:**

  - **Pink pulseless hand** (perfused, no palpable pulse) after reduction: admit, elevate and observe closely. Explore only if perfusion deteriorates or there is **associated median nerve palsy** (the artery may be tethered or entrapped).
  - **White pulseless hand** after closed reduction: **urgent open exploration** of the brachial artery (anterior approach).
  - **Compartment syndrome** (Volkmann ischaemia) risk is higher with ipsilateral forearm fracture (floating elbow) and with **casting in more than 90 degrees of flexion**.
- **Assessing reduction: Baumann angle** (shaft axis vs capitellar physis on AP; about 70-75 degrees, compare with the other side; change over 5 degrees = coronal malalignment), anterior humeral line, **teardrop**, medial and lateral column integrity.
- **Cubitus varus ("gunstock")** is the commonest late complication. It is due to **malunion (medial column collapse, internal rotation)**, not growth disturbance. It is cosmetic but can cause posterolateral rotatory instability, ulnar nerve symptoms and snapping triceps. Treat with a **lateral closing-wedge (French) osteotomy** or dome osteotomy.

Lateral condyle fracture

- Second commonest elbow fracture (age 5-10). It is **intra-articular and physeal (behaves like SH IV)**, pulled by the extensor muscles, so it displaces and has a high **non-union** risk.
- **Milch I** (lateral to the trochlear groove, SH IV, stable elbow) vs **Milch II** (into the trochlear groove, SH II-like, elbow unstable). **Jakob/Weiss** grade by displacement: under 2 mm, 2-4 mm with intact cartilage hinge, over 4 mm and rotated.
- **Internal oblique view** shows the true displacement best. An **arthrogram or MRI** shows whether the articular cartilage

hinge is intact.

- **Treatment:** under 2 mm, cast with weekly radiographs (it can displace); **over 2 mm, ORIF** (posterior dissection is avoided: the blood supply enters posteriorly, so dissecting there causes **AVN of the condyle**).
- **Complications:** non-union, leading to **cubitus valgus** and **tardy ulnar nerve palsy** years later; lateral spur (overgrowth, cosmetic "pseudo-varus"); fishtail deformity.

Medial epicondyle fracture

- Ages 9-14; a traction (apophyseal) injury; about half are associated with **elbow dislocation**.
- **Absolute indication for surgery: incarceration of the fragment in the joint** (after dislocation reduction). Relative: ulnar nerve dysfunction, valgus instability in throwers or gymnasts, displacement in the dominant arm of an athlete.
- Most displaced fractures do well non-operatively (fibrous union is usually painless).

Radial neck, Monteggia and other forearm injuries

- **Radial neck:** angulation **under 30 degrees** is accepted; 30-60 degrees, closed reduction (Patterson manoeuvre, Israeli technique); over 60 degrees or translated, percutaneous joystick or **intramedullary (Metaizeau) reduction**. Avoid open reduction where possible: **loss of forearm rotation (stiffness) is the commonest complication**, and AVN of the radial head and radioulnar synostosis follow open reduction.
- **Monteggia (Bado):** I anterior radial head dislocation with apex-anterior ulnar fracture (**commonest in children**), II posterior, III lateral (often ulnar metaphyseal greenstick), IV anterior dislocation with both-bone fracture. In children, **restoring ulnar length and alignment usually reduces the radial head**; plastic deformation of the ulna must be corrected.

A **missed Monteggia** (chronic) needs ulnar osteotomy with annular ligament reconstruction.

- **Galeazzi:** radial shaft fracture with distal radioulnar joint disruption; the paediatric equivalent is a distal ulnar physeal separation.
- **Forearm shaft:** acceptable alignment is generous in children under about 9 (angulation up to about 15 degrees, bayonet apposition, malrotation up to about 45 degrees); in older children less (about 10 degrees angulation, 30 degrees malrotation). Mid-shaft radius malunion limits rotation most. **Elastic stable intramedullary nails** for unstable fractures in older children. **Refracture** is common (about 5%) after greenstick fractures: complete the greenstick or cast long enough.
- **Distal radius:** the commonest paediatric fracture overall. Huge remodelling capacity: in a child with 2 years of growth left, up to about 20-25 degrees of sagittal angulation is acceptable. **Do not re-manipulate a physeal (SH II) fracture after 5-7 days** (physeal damage).
- **Nursemaid's elbow (pulled elbow):** annular ligament subluxation in children aged 1-4. Reduce by supination with flexion or by hyperpronation (more successful).

4. Proximal humerus and clavicle

- Proximal humerus physeal fractures (SH II in adolescents, SH I in younger children) remodel enormously (up to about 60 degrees accepted in young children). Operate only in adolescents with severe displacement or soft-tissue interposition (biceps tendon).
- **Neonatal clavicle fracture** (difficult delivery, pseudoparalysis): heals without treatment. Differentiate from **obstetric brachial plexus palsy** and **congenital**

pseudarthrosis of the clavicle (right side, painless, no callus, subclavian artery pressure theory).

- **Medial clavicle physeal injury** (the last physis to close, about 23-25 years): posterior displacement can compress mediastinal structures. **CT**, then reduce in theatre with a thoracic surgeon available.

5. Spine in children

- **SCIWORA** (spinal cord injury without radiographic abnormality) is due to the elastic paediatric spine; **MRI** shows the lesion.
- Upper cervical injuries predominate under about 8 years (large head, horizontal facets). **Pseudosubluxation of C2 on C3** (up to about 4 mm) is normal; use the **Swischuk line**.
- Use a **back-board with an occipital recess or shoulder roll** in children under about 6 (the large occiput flexes the neck).
- **Chance (flexion-distraction) fracture** with a lap belt: look for associated **bowel injury**.

6. Pelvis and hip

- Pelvic fractures in children often need less fixation (elastic pelvis), but **associated visceral injuries** are frequent. The triradiate cartilage can arrest (shallow acetabulum).
- **Avulsion fractures in adolescent athletes:** ASIS (sartorius), **AIIS (rectus femoris)**, ischial tuberosity (**hamstrings**, the one that may need fixation if displaced over 1.5-2 cm), lesser trochanter (iliopsoas; **in an adult, a lesser trochanter avulsion = pathological fracture until proved otherwise**).
- **Delbet classification of paediatric hip fractures:**

Type	Site	Frequency	AVN risk
I	Transepiphyseal (with or without dislocation)	Rare	Highest, near 100% with dislocation
II	**Transcervical**	**Commonest (about 45-50%)**	About 50%
III	Cervicotrochanteric (basicervical)	About 30%	About 25%
IV	Intertrochanteric	About 10-15%	Lowest, about 10%

- Treat as an emergency: **urgent anatomical reduction**, stable fixation (avoid crossing the physis if possible in young children; stability is more important than sparing it), and **capsular decompression**. Complications: **AVN (commonest and most serious)**, coxa vara, non-union, premature physeal closure.

7. Femoral shaft

Age / size	Treatment
0-6 months	**Pavlik harness** or posterior splint
6 months-5 years	**Immediate (or early) hip spica**; accept up to about 2 cm shortening, about 15 degrees varus/valgus and about 20 degrees anterior/posterior angulation (less with age)
5-11 years	**Flexible (titanium) elastic nails** for length-stable fractures in children **under about 45-49 kg**; submuscular bridge plating or external fixation for length-unstable or comminuted fractures

Age / size	Treatment
Over about 11 years or heavier	**Rigid intramedullary nail through a lateral trochanteric entry**. Avoid piriformis entry: the lateral ascending cervical branches of the **medial femoral circumflex artery** pass there, so it risks **AVN of the femoral head**

- **Elastic nails:** two nails, each about **40% of the medullary canal diameter** (combined about 80%), pre-bent to the same curve so they balance. **Complications:** loss of length in unstable patterns, nail-end irritation, malunion in heavier children.
- A **femoral fracture in a child who is not yet walking** (under about 1 year) is a red flag for **non-accidental injury**.
- **Distal femoral physeal fractures** (often SH II) have a high rate of growth arrest (about 40-50%) even when well treated: warn the family and follow up for 1-2 years.

8. Knee

- **Tibial spine (eminence) fracture:** ages 8-14, the paediatric ACL equivalent. **Meyers-McKeever:** I undisplaced, II anterior hinge, III completely displaced (IIIA displaced, IIIB rotated), IV comminuted (Zaricznyj). The **intermeniscal ligament or anterior horn of the medial meniscus** can block reduction. Type I and reducible II: long-leg cast in extension. Irreducible II, III and IV: **arthroscopic or open reduction and fixation** (suture or screw).
- **Tibial tubercle avulsion:** adolescent boys jumping; Ogden classification. Watch for **compartment syndrome** (anterior tibial recurrent artery).
- **Patellar sleeve fracture:** a cartilage sleeve pulled off the inferior pole (ages 8-12); the radiograph shows only a small

fleck of bone and patella alta, so the injury is underestimated. ORIF.

- **Proximal tibial metaphyseal fracture (Cozen):** a seemingly innocent undisplaced fracture in a child aged 3-6 develops **progressive genu valgum** over the next year from asymmetric overgrowth. Warn the parents; most remodel; avoid early osteotomy.
- **Proximal tibial physeal (SH) fractures:** risk of **popliteal artery injury** (the artery is tethered at the trifurcation).

9. Tibia and ankle

- **Toddler's fracture:** undisplaced spiral fracture of the distal tibia in a child just walking (9 months-3 years). May only show on an oblique view or at 10-14 days. Usually accidental.
- **Transitional fractures** occur in adolescents, because the distal tibial physis closes over about 18 months: **central first, then medial, then lateral**.
  - **Juvenile Tillaux:** SH III avulsion of the **anterolateral epiphysis** by the **anterior inferior tibiofibular ligament**, as only the lateral physis is still open.
  - **Triplane:** SH IV in three planes. It looks like SH III on the AP view and SH II on the lateral view; two-, three- or four-part types.
  - **CT** for both. **Articular displacement over 2 mm** needs reduction and fixation (screws parallel to the joint, avoiding the physis if growth remains).
- **Distal tibial SH III/IV (medial malleolus):** high risk of physeal bar and **varus** deformity. Anatomical ORIF.

10. Non-accidental injury (NAI)

- Red flags: a history inconsistent with the injury or the child's development, delay in presentation, injuries of different ages, and **any fracture in a non-ambulant infant**.
- **High-specificity fractures: classic metaphyseal lesions** (corner or bucket-handle fractures, from shaking or twisting), **posterior rib fractures** (squeezing), scapula, spinous process and sternum fractures.
- Moderate specificity: multiple or bilateral fractures, fractures of different ages, epiphyseal separations, vertebral body fractures, digital fractures in non-ambulant children, complex skull fractures.
- **Commonest fracture in abuse overall: diaphyseal long-bone fracture** (humerus or femur in infants); **humeral shaft fracture under 3 years** is suspicious.
- **Workup:** full **skeletal survey** in children under 2 (repeat at about 2 weeks), ophthalmology (retinal haemorrhages), head imaging, bloods (bone profile, vitamin D, coagulation). Differentials: **osteogenesis imperfecta**, rickets, scurvy, Caffey disease, congenital syphilis, normal metaphyseal variants.
- The treating doctor has a duty to **report concerns** (in India: child protection services; POCSO where sexual abuse is suspected) and to admit the child for safety.

> Exam pearl: **AIN is the commonest nerve injured in a supracondylar fracture; the ulnar nerve is the commonest iatrogenic injury (medial pin).** A **lateral condyle fracture over 2 mm displaced needs ORIF**, and its late complication is **tardy ulnar palsy from cubitus valgus**.

Exam pearl: **Delbet II is the commonest paediatric hip fracture; Delbet I has the highest AVN risk. Piriformis-entry nails are avoided in children because of AVN.**

cerebral palsy and neuromuscular

1. Cerebral palsy: definition and classification

- A group of permanent disorders of movement and posture caused by a **non-progressive lesion of the immature brain** (before, during or within about 2 years after birth). The brain lesion is static, but the **musculoskeletal problems are progressive** (muscles do not grow at the rate of bone under spasticity, so contractures and bony deformity develop with growth).
- **Risk factors:** prematurity and low birth weight (strongest), perinatal hypoxia, intrauterine infection, multiple pregnancy, kernicterus (athetoid CP), neonatal stroke.
- **Physiological classification:**
  - **Spastic** (about 70-80%; upper motor neurone, velocity-dependent tone; pyramidal tract).
  - **Dyskinetic** (athetosis, dystonia, chorea; basal ganglia; kernicterus). **Poor candidates for tendon surgery** (unpredictable results).
  - **Ataxic** (cerebellum).
  - Mixed.
- **Topographic classification: hemiplegia** (arm worse than leg; walks by about 18-24 months; seizures common), **diplegia** (legs much worse than arms; classic for prematurity with **periventricular leukomalacia**), **quadriplegia / total body involvement** (all limbs, poor head control, intellectual disability, seizures, feeding problems).
- **GMFCS (Gross Motor Function Classification System), age 6-12 descriptors:**

Level	Function
I	Walks without limitation; difficulty only with advanced skills
II	Walks with limitations (long distances, uneven ground, stairs need a rail)
III	Walks with a hand-held mobility device (walker, crutches); wheelchair for long distances
IV	Self-mobility limited; powered mobility; may walk short distances with assistance
V	Transported in a manual wheelchair; very limited head and trunk control

- GMFCS is stable over time and predicts **hip displacement risk** (about 0% at level I to about 90% at level V) and walking prognosis.
- **Walking prognosis:** sitting independently by **age 2** predicts walking; not sitting by 4 suggests the child will not walk. Persistent primitive reflexes after age 1 (Bleck's criteria) are poor signs.

2. Principles of management

- **Team approach:** physiotherapy, orthoses, tone management, orthopaedic surgery. Goals depend on GMFCS: improve gait in I-III; keep sitting comfortable, hips painless, and spine balanced in IV-V.
- **Tone management:**
  - **Botulinum toxin A** (blocks acetylcholine release at the neuromuscular junction) for focal **dynamic** spasticity in young children (typically 2-6), effect lasts about 3-6 months. Combine with serial casting. Not useful for fixed contracture.
  - **Oral baclofen** (GABA-B agonist; sedation limits use); **intrathecal baclofen pump** for severe generalised

spasticity or dystonia (GMFCS IV-V).
    - **Selective dorsal rhizotomy (SDR):** sectioning a proportion (often about half) of the L2-S1 dorsal rootlets, guided by intraoperative stimulation. Ideal candidate: **spastic diplegia, GMFCS II-III, age about 4-8**, good strength and selective motor control, minimal fixed contractures, no dystonia. Permanent reduction of spasticity; may unmask weakness; later hip subluxation and spinal deformity can occur.
- **Gait analysis** (3D kinematics, kinetics, EMG) before multilevel surgery.
- **Single-event multilevel surgery (SEMLS):** correct all soft-tissue and bony deformities in one operation (rather than "birthday syndrome" of one operation a year), typically at **age about 6-10** when gait has matured. Followed by intensive rehabilitation for about a year.

3. The hip in CP

- **Spastic hip displacement** is due to overactive **adductors and iliopsoas** with weak abductors and extensors, leading to **coxa valga, increased femoral anteversion, and acetabular dysplasia (posterosuperior)**. The hip subluxes **posterosuperiorly** (unlike DDH).
- **Hip surveillance:** clinical exam and AP pelvis radiograph from about age 2, every 6-12 months, more often for GMFCS III-V (Australian/CPUP-based protocols).
- **Reimers migration percentage (MP):** percentage of the femoral head width lateral to Perkin's line. Normal under about 10%; **over 30-40% = subluxation**; 100% = dislocation. **Progression over about 10% a year** is significant.
- **Management by severity:**

Situation	Surgery
Young child (under about 6-8), **MP 30-40%** (some use 25-60%)	**Preventive soft-tissue release:** adductor longus and gracilis tenotomy ± iliopsoas lengthening (psoas over brim)
MP over 40-50%, older child, or failed release	**Reconstructive: varus derotation osteotomy (VDRO)** of the proximal femur ± **pelvic osteotomy (Dega or San Diego)** for acetabular dysplasia, with open reduction if needed
Painful, long-standing dislocation with arthritic head in a non-walker	**Salvage:** proximal femoral resection (Castle or McHale), valgus osteotomy (Schanz), arthrodesis or arthroplasty

- Do not do an isolated obturator neurectomy (anterior branch) routinely; abandoned by most because of over-abduction (windswept or abduction contracture).
- **Windswept hips:** one hip abducted, the other adducted and dislocated, with pelvic obliquity and scoliosis.

4. The knee in CP

- **Crouch gait** (excessive knee flexion in stance): due to hamstring contracture, weak soleus or plantarflexor-knee-extension couple insufficiency (often **after isolated over-lengthening of the Achilles tendon**), lever-arm dysfunction (external tibial torsion, foot instability). Treat with hamstring lengthening, correction of lever arms, **distal femoral extension osteotomy** with **patellar tendon advancement** in severe fixed crouch, and floor-reaction AFOs.
- **Stiff-knee gait:** reduced and delayed knee flexion in swing from an **overactive rectus femoris** (confirmed by dynamic

EMG; Duncan-Ely test). Treat with **distal rectus femoris transfer** (to the semitendinosus or gracilis) rather than release.

- **Hamstring tightness:** popliteal angle increased. Medial hamstring lengthening; lateral lengthening only if needed (risk of recurvatum and crouch). **Hamstring lengthening increases anterior pelvic tilt** (the hamstrings are hip extensors too).
- **Jump gait** (equinus + knee and hip flexion) and **true equinus** (hemiplegia). The Rodda and Graham sagittal gait classification: true equinus, jump, apparent equinus, crouch.

5. The foot in CP

- **Equinus** is the commonest deformity.
  - **Silfverskiold test:** if dorsiflexion improves with the knee flexed, the **gastrocnemius alone** is tight: **gastrocnemius recession** (Strayer, Vulpius, Baker). If not, both are tight: **tendo-Achilles lengthening** (Hoke triple hemisection, Z-lengthening).
  - **Avoid over-lengthening the Achilles tendon** in diplegia: it causes **crouch gait**, which is far worse than mild equinus. Diplegic equinus is often "apparent" (from knee and hip flexion).
- **Equinovarus** (hemiplegia): overactive **tibialis posterior** (split posterior tibial tendon transfer, SPLATT-type to peroneus brevis, or lengthening) or **tibialis anterior** (split anterior tibial tendon transfer, **SPLATT**). Confirm the culprit with dynamic EMG.
- **Equinovalgus / planovalgus** (diplegia and quadriplegia): peroneal overactivity and gastrocnemius-soleus contracture. **Calcaneal lengthening osteotomy (Evans)** in mobile feet; **subtalar or triple arthrodesis** in rigid or severe feet in older children.

- **Hallux valgus** develops with planovalgus: first MTP arthrodesis.

6. Spine and upper limb in CP

- **Neuromuscular scoliosis:** long **C-shaped thoracolumbar curves** with **pelvic obliquity**, mainly in GMFCS IV-V. Progresses even after skeletal maturity (especially curves over about 40 degrees).
- **Posterior spinal fusion from T2 to the pelvis** (pelvic fixation with iliac or **S2-alar-iliac** screws) when the curve exceeds about 40-50 degrees and sitting balance worsens. Complications are higher than in idiopathic scoliosis: infection, blood loss, pulmonary problems. Optimise nutrition first.
- **Upper limb (hemiplegia):** shoulder internal rotation, elbow flexion, forearm pronation, wrist and finger flexion, **thumb-in-palm**. Surgery (e.g. **flexor carpi ulnaris to ECRB transfer**, pronator teres rerouting, thumb adductor release) mainly improves function and hygiene; needs good sensation and voluntary control (the House and Zancolli classifications).

7. Muscular dystrophies

Duchenne muscular dystrophy (DMD)

- **X-linked recessive**, absent **dystrophin** (Xp21; deletions in about 65%); one-third are new mutations. Boys; incidence about 1 in 3,500-5,000 male births.
- Presents at **3-5 years**: clumsiness, frequent falls, **toe-walking**, **Gowers sign** (climbing up the thighs to stand: proximal weakness), **calf pseudohypertrophy** (fat and fibrous replacement), waddling gait, lumbar lordosis.
- **CK massively raised** (10-100 times normal, even at birth); diagnosis by **genetic testing** (multiplex ligation-dependent

probe amplification); muscle biopsy (absent dystrophin on immunostaining) if needed.

- **Natural history:** loss of walking around **age 12** (later on steroids), scoliosis after loss of ambulation, death in the 20s-30s from **cardiomyopathy** or respiratory failure.
- **Corticosteroids** (prednisolone or deflazacort) prolong walking by about 2-3 years, reduce scoliosis and improve cardiorespiratory function. Exon-skipping drugs (e.g. eteplirsen for exon 51) and gene therapy for selected mutations.
- **Orthopaedics:** keep walking as long as possible (Achilles and ITB releases with KAFOs in selected cases). **Scoliosis:** fuse **early**, when the curve is **about 20-30 degrees**, before the **FVC falls below about 30-35%** (anaesthetic risk rises sharply). Long fusion to the pelvis or L5. **Malignant hyperthermia-like reactions** and rhabdomyolysis: avoid succinylcholine and volatile agents.
- Fractures from osteoporosis (steroids, immobility): treat quickly to keep walking.

Becker and other dystrophies

- **Becker:** reduced or abnormal dystrophin; later onset (over about 7), walks beyond 16, milder, cardiomyopathy common.
- **Limb-girdle** (autosomal), **facioscapulohumeral** (AD; scapular winging, facial weakness: **scapulothoracic fusion** for winging), **Emery-Dreifuss** (early **elbow flexion contractures**, Achilles contracture, **cardiac conduction block**: pacemaker).
- **Myotonic dystrophy** (CTG repeat; distal weakness, myotonia, cataracts, frontal balding, cardiac conduction defects).

8. Spinal muscular atrophy (SMA)

- Autosomal recessive **SMN1 deletion**, loss of **anterior horn cells**: symmetric proximal weakness, absent reflexes, tongue fasciculation, **normal sensation and intelligence**. SMN2 copy number modifies severity.
- **Types:** I (Werdnig-Hoffmann; never sits, death in infancy without treatment), II (sits but never stands), III (Kugelberg-Welander; walks), IV adult.
- **Nusinersen** (antisense oligonucleotide, intrathecal), **onasemnogene abeparvovec** (gene therapy) and **risdiplam** (oral) have transformed survival.
- **Scoliosis** almost universal in type II: bracing for sitting, then **growth-friendly instrumentation** (growing rods, magnetically controlled) or fusion; leave access for intrathecal therapy if possible.
- **Hip subluxation and dislocation** are common but usually **painless: leave them alone**.

9. Myelomeningocele (spina bifida)

- Folic acid deficiency, valproate. Associated **Arnold-Chiari II malformation, hydrocephalus (shunt)**, tethered cord, syringomyelia, neurogenic bladder, **latex allergy** (latex-free theatre).
- **Motor level determines function:**

Level	Key muscle working	Typical function
Thoracic / L1-L2	Hip flexors only	Wheelchair; standing in a frame

Level	Key muscle working	Typical function
L3-L4	**Quadriceps (L3-4), tibialis anterior (L4)**	Household or community walker with AFOs/crutches; **functional quadriceps is the key to walking**
L5	Hamstrings, toe extensors, gluteus medius partly	Community walker, AFOs
Sacral	Gastrocnemius-soleus	Near-normal walking

- **Hip dislocation** is commonest at L3-L4 (unopposed hip flexors and adductors vs weak abductors and extensors). **Do not reduce** a dislocated hip in most cases; correct contractures to keep the pelvis level.
- **Rapidly increasing scoliosis, new spasticity, deteriorating gait, changing foot deformity or urinary function:** suspect **tethered cord, shunt malfunction or syrinx**; MRI and neurosurgical review.
- **Feet:** aim for a **plantigrade, braceable foot**. Rigid clubfoot (talectomy in severe cases), vertical talus, calcaneus foot (L4-L5; anterior tibial transfer to the calcaneus). **Insensate feet** develop pressure ulcers and Charcot changes.
- **Fractures** in insensate limbs present as warm, swollen limbs with fever (mistaken for infection) and heal with exuberant callus.

10. Hereditary neuropathies and other conditions

- **Charcot-Marie-Tooth (HMSN):** commonest inherited neuropathy. **Type 1A: PMP22 duplication on chromosome 17**, demyelinating, slow conduction velocities. Type 2: axonal.
  - **Cavovarus foot**, claw toes, peroneal muscle wasting ("**inverted champagne bottle**" legs), intrinsic hand

weakness.
    - **Mechanism:** weak **tibialis anterior** is overpowered by **peroneus longus** (plantarflexed first ray, forefoot-driven cavus); weak **peroneus brevis** is overpowered by **tibialis posterior** (hindfoot varus).
    - **Coleman block test:** a block under the lateral forefoot lets the first ray drop; if the **hindfoot varus corrects**, the hindfoot is flexible and the deformity is forefoot driven: plantar fascia release, **first metatarsal dorsiflexion osteotomy**, **peroneus longus to brevis transfer**, ± tibialis posterior transfer. If not, add a **lateralising calcaneal osteotomy**; triple arthrodesis only for rigid, arthritic feet.
- **Friedreich ataxia:** AR, **GAA trinucleotide repeat in frataxin** (chromosome 9). Ataxia from about age 10, areflexia, cavovarus feet, scoliosis (more like idiopathic in shape), **hypertrophic cardiomyopathy** (the main cause of death), diabetes.
- **Arthrogryposis multiplex congenita:** multiple **non-progressive** congenital contractures, normal sensation and usually normal intelligence. **Amyoplasia** (classic): internally rotated shoulders, extended elbows, flexed wrists, rigid **clubfeet**, dislocated hips, knee contractures. Early stretching; surgery aims for walking and independent function (bilateral hip dislocations are usually left; unilateral may be reduced).
- **Poliomyelitis (residual paralysis)** is still examined in India: asymmetric flaccid paralysis with normal sensation; tendon transfers once power has plateaued (about 2 years), "**a muscle loses one grade of power after transfer**", and transfer **in phase** where possible.

Exam pearl: **Silfverskiold test positive = gastrocnemius recession; over-lengthening the Achilles in a diplegic child causes crouch gait. Stiff-knee gait = distal rectus femoris transfer.**

Exam pearl: **DMD scoliosis: fuse early (curve about 20-30 degrees, FVC above about 30-35%). SMA hip dislocation: leave alone. Myelomeningocele: L3-L4 (quadriceps) is the key level for walking.**

skeletal dysplasias

1. Approach to a short child

- **Proportionate short stature:** constitutional, familial, endocrine (hypothyroidism, growth hormone deficiency), nutritional, systemic disease.
- **Disproportionate short stature = skeletal dysplasia** until proved otherwise:
  - **Short-limbed** (achondroplasia, pseudoachondroplasia, metaphyseal dysplasias) vs **short-trunk** (spondyloepiphyseal dysplasias, Morquio).
  - Pattern of limb shortening: **rhizomelic** (proximal: humerus and femur; achondroplasia), **mesomelic** (forearm and leg; Langer, dyschondrosteosis/Leri-Weill with **Madelung deformity**, SHOX gene), **acromelic** (hands and feet).
- Radiographic approach: which part of the bone is affected? **Epiphyseal** (MED, SED), **metaphyseal** (achondroplasia, metaphyseal chondrodysplasias, rickets), **diaphyseal** (Camurati-Engelmann), and is the **spine** involved (spondylo-)?
- **Upper cervical instability screen** (flexion-extension films ± MRI) before anaesthesia in **Morquio, SED congenita, pseudoachondroplasia, Down syndrome, Larsen syndrome, metatropic dysplasia**.

2. FGFR3 group

- **FGFR3 is a negative regulator of chondrocyte proliferation:** activating (gain-of-function) mutations slow growth at the physis.
- **Achondroplasia** (commonest skeletal dysplasia, about 1 in 25,000):

  - **Autosomal dominant**, **FGFR3 G380R** (Gly380Arg) in about 98%; **about 80% are new mutations** associated with **advanced paternal age**. Homozygous achondroplasia is lethal.
  - **Clinical:** rhizomelic shortening, **frontal bossing**, midface hypoplasia, **trident hand** (separation of middle and ring fingers), **genu varum** (fibular overgrowth), thoracolumbar kyphosis in infancy (usually resolves when walking), lumbar hyperlordosis, normal trunk length, normal intelligence. Average adult height about 125-130 cm.
  - **Radiographs: interpedicular distance narrows from L1 to L5** (normally widens), short pedicles, posterior vertebral scalloping, **champagne-glass pelvic inlet**, squared iliac wings, flat acetabular roofs, inverted-V physes at the knee.
  - **Spine problems: foramen magnum stenosis** in infancy (central apnoea, hypotonia, **sudden death**; screen with MRI/polysomnography; suboccipital decompression); **thoracolumbar kyphosis** (avoid unsupported sitting; bracing; fusion if progressive over about 30 degrees after age 3-5); **lumbar spinal stenosis in adults** (neurogenic claudication; wide laminectomy and fusion).
  - **Genu varum:** osteotomy or guided growth if symptomatic. Limb lengthening is controversial.
  - **Vosoritide** (C-type natriuretic peptide analogue; antagonises FGFR3 signalling) increases annual growth velocity.
- **Hypochondroplasia** (FGFR3, milder; normal face). **Thanatophoric dysplasia** (FGFR3; lethal; "telephone-receiver" femora, cloverleaf skull).

3. COMP and type IX collagen group

- **Pseudoachondroplasia** (COMP, cartilage oligomeric matrix protein; AD):

  - **Normal face and normal length at birth**; short limbs become apparent at **2-3 years**.
  - Marked **ligamentous laxity**, windswept knees, **odontoid hypoplasia with C1-C2 instability**, small irregular epiphyses, flared metaphyses, early **osteoarthritis** of hips and knees.
  - Distinguish from achondroplasia: normal face and skull, normal interpedicular distance.
- **Multiple epiphyseal dysplasia (MED)** (COMP, type IX collagen COL9A1-3, matrilin-3; usually AD):
  - Mild short stature, joint pain and stiffness, **small, irregular, flattened epiphyses**, **double-layered patella** (recessive type, SLC26A2).
  - **Bilateral, symmetrical femoral head changes at the same stage**, normal spine, acetabular involvement: differentiates it from **bilateral Perthes** (asynchronous, different stages).
  - **Premature osteoarthritis**: hip arthroplasty in early adulthood.

4. Type II collagen group (COL2A1)

- Type II collagen is in **articular cartilage, physis, vitreous humour and the inner ear**, which explains the eye and hearing problems.
- **Spondyloepiphyseal dysplasia (SED) congenita** (AD, COL2A1):
  - **Short-trunk** dwarfism present at birth, barrel chest, **coxa vara**, **odontoid hypoplasia / os odontoideum with atlantoaxial instability** (fuse C1-C2 if instability or myelopathy), kyphoscoliosis.

  - **Myopia and retinal detachment** (annual eye examination), hearing loss, cleft palate.
  - Delayed epiphyseal ossification (femoral heads may not be seen for years); early OA.
- **SED tarda:** X-linked (TRAPPC2/SEDL), boys, late childhood, hip and back pain, platyspondyly with a "heaped-up" posterior vertebral body.
- **Kniest dysplasia** (COL2A1): **dumbbell-shaped femora**, large stiff joints, myopia and **retinal detachment**, hearing loss, cleft palate.
- **Stickler syndrome** (COL2A1, COL11A1): myopia and retinal detachment, Pierre Robin sequence, early OA.

5. Other named dysplasias

- **Diastrophic dysplasia** (AR, **SLC26A2** sulfate transporter, common in Finland):
  - **"Hitchhiker" thumb** (short, proximally placed, abducted first metacarpal), **cauliflower ears** (cystic swelling in infancy), cleft palate.
  - **Rigid clubfeet** (very resistant to treatment), joint contractures, **cervical kyphosis** (can cause quadriplegia: monitor and fuse), scoliosis.
- **Cleidocranial dysplasia** (AD, **RUNX2/CBFA1**, master gene for osteoblast differentiation):
  - **Absent or hypoplastic clavicles** (shoulders can be brought together in front of the chest), **delayed closure of fontanelles** and wide sutures with **wormian bones**, **supernumerary teeth** and delayed tooth eruption, **coxa vara**, wide symphysis pubis.
  - Usually needs no treatment; coxa vara with Hilgenreiner-epiphyseal angle over 60 degrees needs valgus osteotomy.

- **Metaphyseal chondrodysplasias:**

Type	Gene / inheritance	Features
Schmid (commonest)	**COL10A1** (type X collagen), AD	Mild short stature, **coxa vara**, genu varum, waddling gait. **Looks like rickets but calcium, phosphate, ALP and vitamin D are normal**
Jansen	**PTH/PTHrP receptor** (activating), AD	Severe, **hypercalcaemia**, hypophosphataemia
McKusick (cartilage-hair hypoplasia)	**RMRP**, AR (Amish)	Fine sparse hair, **immune deficiency**, Hirschsprung disease, risk of lymphoma
Shwachman-Diamond	SBDS, AR	**Pancreatic insufficiency**, neutropenia

- **Mucopolysaccharidoses** (lysosomal storage, defective glycosaminoglycan breakdown; urinary GAGs raised):

Type	Name	Inheritance	Key features
I	**Hurler** (alpha-L-iduronidase)	AR	Coarse face, **intellectual disability**, corneal clouding, gibbus (thoracolumbar kyphosis), carpal tunnel syndrome; stem-cell transplant
II	**Hunter** (iduronate sulfatase)	**X-linked recessive** (the only XLR MPS)	Like Hurler, **no corneal clouding**, milder

Type	Name	Inheritance	Key features
IV	**Morquio** (galactose-6-sulfatase; keratan sulfate)	AR	**Normal intelligence**, short trunk, pectus carinatum, **odontoid hypoplasia with atlantoaxial instability** (occipitocervical fusion often needed), ligament laxity, genu valgum, corneal clouding

    - **Radiographs (dysostosis multiplex):** oar-shaped ribs, bullet-shaped phalanges, anterior beaking of lumbar vertebrae (inferior beaking in Hurler, central in Morquio), proximally pointed metacarpals.
- **Osteopetrosis (marble bone disease): failure of osteoclast resorption**.
    - **Infantile malignant** (AR; TCIRG1, CLCN7): marrow failure (anaemia, hepatosplenomegaly), cranial nerve compression (blindness, deafness), early death without **haematopoietic stem cell transplant** (osteoclasts are of haematopoietic origin).
    - **Adult benign (Albers-Schonberg, AD, CLCN7):** fractures, osteomyelitis of the mandible.
    - **Carbonic anhydrase II deficiency:** with **renal tubular acidosis** and cerebral calcification.
    - **Radiographs:** dense, brittle bones, **bone-in-bone**, **Erlenmeyer flask** distal femora, **rugger-jersey spine**, sandwich vertebrae.
    - Fractures heal slowly; **drilling and fixation are difficult** (very hard bone; use cooled drills and plates rather than nails).
- **Pyknodysostosis** (AR, **cathepsin K**): dense bones, short stature, **acro-osteolysis** of terminal phalanges, open

fontanelles, obtuse mandibular angle, fractures. (Toulouse-Lautrec is thought to have had it.)

- **Camurati-Engelmann** (TGF-beta 1): symmetric **diaphyseal** cortical thickening, leg pain, waddling gait.
- **Melorheostosis:** "dripping candle-wax" cortical hyperostosis along one sclerotome; pain and contractures.
- **Osteopoikilosis** (multiple small sclerotic islands near joints, asymptomatic) and **osteopathia striata** (linear striations): incidental findings.

6. Osteogenesis imperfecta (OI)

- About **90% are COL1A1 or COL1A2 mutations** (type I collagen, quantity or quality), usually AD; recessive forms involve collagen-processing genes (CRTAP, LEPRE1/P3H1, SERPINF1 etc.).
- **Sillence classification:**

Type	Severity	Sclerae	Notes
I	Mild, non-deforming	**Blue**	Commonest; fractures decrease after puberty; **hearing loss** (third decade)
II	**Perinatal lethal**	Blue	Crumpled long bones, beaded ribs
III	**Most severe in survivors**, progressive deformity	Blue at birth, whiter with age	Very short, wheelchair-dependent, **dentinogenesis imperfecta**, triangular face
IV	Moderate	**Normal (white)**	Variable deformity

Type	Severity	Sclerae	Notes
V	Moderate	Normal	**Hyperplastic callus**, **calcification of the interosseous membrane** of the forearm (limits rotation), radial head dislocation; IFITM5; normal collagen

- **Other features: wormian bones** in the skull, ligament laxity, easy bruising, basilar invagination, scoliosis (common, progressive; bracing ineffective; fuse curves over about 45-50 degrees), **dentinogenesis imperfecta**, hyperthermia under anaesthesia.
- **Bisphosphonates** (cyclical IV **pamidronate or zoledronate**) increase bone density and **reduce fracture rate and pain**; they do not cure the collagen defect. **Denosumab** has been tried (rebound hypercalcaemia on stopping).
- **Surgery:** correct long-bone bowing with **multiple osteotomies and intramedullary fixation** (**Sofield-Millar "shish-kebab" osteotomies**). **Telescoping rods** that elongate with growth (**Fassier-Duval**, Bailey-Dubow, Sheffield) reduce reoperation. **Avoid plates** (stress risers, peri-plate fractures).
- **Differential: non-accidental injury.** OI is suggested by blue sclerae, wormian bones, osteopenia, dentinogenesis imperfecta, family history, and fractures in hospital.

7. Neurofibromatosis type 1 (NF1)

- AD, **neurofibromin** (chromosome 17), a tumour suppressor (Ras-GTPase activating protein). Half are new mutations.
- **Diagnostic criteria (2 or more):** 6 or more **cafe-au-lait spots** (over 5 mm before puberty, over 15 mm after), 2 or more

neurofibromas or one plexiform neurofibroma, **axillary or inguinal freckling**, optic glioma, 2 or more **Lisch nodules** (iris hamartomas), a distinctive osseous lesion (**sphenoid wing dysplasia or long-bone cortical thinning / tibial pseudarthrosis**), a first-degree relative with NF1.

- **Orthopaedic problems:**
  - **Dystrophic scoliosis: short, sharp, angular** curves (often kyphoscoliosis), **rib penciling**, vertebral scalloping, enlarged foramina, dural ectasia. **Rapid progression**; bracing fails; early **combined anterior and posterior fusion** (high pseudarthrosis rate). Non-dystrophic curves behave like idiopathic scoliosis but can "modulate" into dystrophic curves.
  - **Congenital pseudarthrosis of the tibia: anterolateral bowing** of the tibia in infancy (about half have NF1). **Protect with a total contact orthosis to prevent fracture**; avoid osteotomy for the bow. Once a pseudarthrosis develops: excision with **intramedullary rodding and bone grafting**, **Ilizarov bone transport**, or **vascularised fibular graft**; bone morphogenetic protein and cross-union techniques. Persistent non-union may end in amputation. (Compare: **posteromedial bowing** is benign and associated with calcaneovalgus foot and leg-length discrepancy; **anteromedial bowing** goes with fibular hemimelia.)
  - **Malignant peripheral nerve sheath tumour** (about 10% lifetime risk; a painful, enlarging plexiform neurofibroma needs MRI and PET).
  - **Hemihypertrophy**, overgrowth (elephantiasis neuromatosa), non-ossifying fibromas.

8. Tumour-like dysplasias

- **Hereditary multiple exostoses (multiple osteochondromas):** AD, **EXT1 (chromosome 8) or EXT2 (chromosome 11)**.
  - Sessile and pedunculated osteochondromas pointing away from the joint; short stature, **forearm deformity** (**ulnar shortening**, radial bowing, radial head dislocation), genu valgum, ankle valgus.
  - **Malignant transformation to secondary chondrosarcoma** in about 1-5% (much higher than a solitary osteochondroma, under 1%). Warning signs: **pain or growth after skeletal maturity**, **cartilage cap over 2 cm** in adults (MRI).
- **Ollier disease** (multiple **enchondromatosis**, unilateral predominance, non-hereditary; IDH1/IDH2 mutations): limb shortening and angular deformity; chondrosarcoma risk about 25-40%.
- **Maffucci syndrome** (enchondromas + **soft-tissue haemangiomas**, phleboliths): higher malignancy risk, including **non-skeletal malignancies** (gliomas, gastrointestinal cancers).
- **Fibrous dysplasia** (**GNAS1** activating mutation, postzygotic): polyostotic forms cause the **shepherd's crook** proximal femur. **McCune-Albright syndrome:** polyostotic fibrous dysplasia, **cafe-au-lait spots with "coast of Maine" borders**, precocious puberty.
- **Fibrodysplasia ossificans progressiva: ACVR1** mutation, **short great toes (hallux valgus) at birth**, progressive heterotopic ossification along fascial planes. **Never biopsy or operate** (flare-ups).

9. Chromosomal and connective tissue conditions with orthopaedic features

- **Down syndrome (trisomy 21): atlantoaxial instability** (atlanto-dens interval over about 5 mm), ligament laxity, **hip instability (dislocation in later childhood)**, **patellofemoral instability**, flatfoot, slipped epiphysis, scoliosis.
- **Marfan syndrome** (FBN1, fibrillin-1; AD): tall, arachnodactyly (**Steinberg thumb sign**, **Walker-Murdoch wrist sign**), **upward lens dislocation**, **aortic root dilatation**, scoliosis (larger, more rigid, higher complication rates), **protrusio acetabuli**, **dural ectasia**, pectus.
- **Homocystinuria** (cystathionine beta-synthase, AR): marfanoid habitus with **downward lens dislocation**, **intellectual disability**, osteoporosis and **thrombosis** (anaesthetic risk).
- **Ehlers-Danlos syndrome:** joint hypermobility, skin hyperextensibility; vascular type (COL3A1) has arterial rupture; avoid stabilisation surgery where possible (poor results).
- **Larsen syndrome** (filamin B): multiple congenital dislocations (knees, hips, elbows), cervical kyphosis, flat face.
- **Turner syndrome:** short stature, **cubitus valgus**, short fourth metacarpal, **Madelung deformity**.

> Exam pearl: **FGFR3 = achondroplasia (interpedicular distance narrows caudally, foramen magnum stenosis). COMP = pseudoachondroplasia and MED. COL2A1 = SED congenita, Kniest, Stickler (eyes!). SLC26A2 = diastrophic (hitchhiker thumb). RUNX2 = cleidocranial. COL10A1 = Schmid. GNAS = fibrous dysplasia. EXT1/2 = HME. ACVR1 = FOP.**

Exam pearl: Screen the **C1-C2** level before anaesthesia in **Morquio, SED, pseudoachondroplasia and Down syndrome**; watch **cervical kyphosis** in diastrophic dysplasia and Larsen syndrome.

bone and joint infection in children

1. Pathophysiology

- **Acute haematogenous osteomyelitis (AHO)** starts in the **metaphysis**: nutrient artery branches end in **sinusoidal loops beside the physis** where blood flow is slow and turbulent, the capillary endothelium has gaps, and there are few phagocytes. Minor trauma may create a haematoma that seeds bacteria.
- **Spread:** pus raises intraosseous pressure, tracks through the **Haversian and Volkmann canals** to the subperiosteal space, and lifts the loose paediatric periosteum (**subperiosteal abscess**). Cortical bone loses its blood supply (from both sides) and dies: **sequestrum**. The elevated periosteum lays down new bone: **involucrum**. Openings in the involucrum through which pus escapes: **cloacae**.
- **Age-related anatomy:**
  - **Under about 18 months: transphyseal vessels** cross the physis, so metaphyseal infection spreads to the **epiphysis and joint** (combined osteomyelitis and septic arthritis; physeal and epiphyseal damage).
  - **Childhood:** the physis is a barrier; infection stays metaphyseal and spreads subperiosteally (thick, loose periosteum).
  - **Adults:** periosteum firmly attached; infection spreads within the medullary canal and into the joint once the physes close.
- **Joints where the metaphysis is intra-articular** (osteomyelitis can rupture into the joint): **proximal femur (hip)**, **proximal humerus (shoulder)**, **proximal radius (elbow)**, **distal lateral tibia (ankle)**.
- **Septic arthritis destroys cartilage quickly:** bacterial toxins and **proteolytic enzymes from neutrophils** (proteoglycan

loss within hours, collagen within days). The **hip is the emergency**: raised intracapsular pressure can tamponade the retinacular vessels and cause **AVN of the femoral head**.

2. Organisms by age and setting

Setting	Organisms
Neonate (under 2 months)	**Group B streptococcus**, **Staphylococcus aureus**, Gram-negative enteric bacilli (E. coli), Candida in NICU babies
6 months-4 years	**S. aureus** and **Kingella kingae** (the commonest cause of osteoarticular infection in this age group where PCR is used)
Older children	**S. aureus** (by far commonest at all ages), group A streptococcus
Adolescents (sexually active)	Neisseria gonorrhoeae (septic arthritis, tenosynovitis, skin pustules)
Sickle cell disease	**Salmonella** is characteristic (bowel infarcts allow bacteraemia), S. aureus also common
Puncture through a sneaker	**Pseudomonas aeruginosa** (osteochondritis of the foot)
Unvaccinated child	**Haemophilus influenzae type b** (now rare)
Cat or dog bite	Pasteurella multocida
Chronic granulomatous disease	Aspergillus, Serratia, Burkholderia
India (endemic)	**Mycobacterium tuberculosis** in subacute cases, dactylitis, monoarthritis

- **MRSA** (community-acquired, especially **Panton-Valentine leukocidin (PVL)-positive** strains): severe sepsis, **multifocal**

disease, **deep vein thrombosis and septic pulmonary emboli**, necrotising myositis, more surgery.

- **Kingella kingae:** Gram-negative coccobacillus in the oropharynx of toddlers, often after a viral upper respiratory infection. **Mild illness, low-grade or no fever, normal or mildly raised CRP and WBC.** It grows poorly on standard media: **inoculate blood-culture bottles** with joint fluid and use **PCR**. Treat with a cephalosporin (resistant to clindamycin, glycopeptides).

3. Clinical features

- **AHO:** fever, malaise, **localised metaphyseal tenderness**, pseudoparalysis (refusal to use the limb), limp; swelling and erythema later.
- **Septic arthritis:** severely painful joint, **refusal to bear weight**, held in the position of maximum capsular volume (**hip: flexion, abduction and external rotation**), pain on any passive movement (log-roll), fever.
- **Neonates: few systemic signs**; irritability, pseudoparalysis, poor feeding; **multifocal** disease is common; high risk of joint destruction.
- **Psoas abscess, pyomyositis, discitis and sacroiliitis** can mimic septic hip: discitis in toddlers presents with refusal to walk or sit; pyomyositis (S. aureus, tropical) with a tender, woody muscle.

4. Investigations

- **Blood:** WBC, **ESR** (rises over days, falls slowly over weeks), **CRP** (rises within 6-8 hours, peaks at 48 hours, **normalises within about a week of effective treatment**: the best marker of response), **blood cultures** (positive in about 30-50%).

- **Kocher criteria (septic arthritis vs transient synovitis of the hip): non-weight-bearing**, **temperature over 38.5 C**, **ESR over 40 mm/h**, **WBC over 12,000/mm3**.

Criteria present	Predicted probability of septic arthritis
0	about 0.2%
1	about 3%
2	about 40%
3	about 93%
4	about 99.6%

  - **Caird** added **CRP over 2 mg/dL** (20 mg/L) as an independent predictor; **fever over 38.5 C is the best single predictor** in that study. Later validation studies showed lower accuracy in other populations: use the criteria as a guide, not a rule.

- **Imaging:**
  - **Plain radiographs:** soft-tissue swelling and loss of fat planes early; **bone changes (lysis, periosteal reaction) appear only after about 10-14 days** (30-50% loss of bone mineral needed). Baseline and to exclude differentials.
  - **Ultrasound: hip effusion** (absence of an effusion makes septic hip very unlikely), subperiosteal collections; guides aspiration.
  - **MRI** is the **investigation of choice for osteomyelitis**: marrow oedema (T1 low, T2/STIR high), subperiosteal and intraosseous abscess, physeal involvement, adjacent septic arthritis, pyomyositis. Contrast helps define abscesses.
  - Bone scan (Tc-99m) if the site is unclear or multifocal disease is suspected; can be cold early in severe infection (compromised blood flow).

- **Aspiration** (joint or subperiosteal pus) under ultrasound or fluoroscopy before antibiotics if it does not delay treatment: Gram stain, culture (plus blood-culture bottle and PCR for Kingella), cell count. **Synovial WBC over about 50,000/mm3 with over 75-90% neutrophils** suggests septic arthritis (lower counts in Kingella and early infection).

5. Treatment

Antibiotics

- Start **empirical IV antibiotics immediately after cultures**: **anti-staphylococcal** cover (cefazolin or cloxacillin/flucloxacillin; **add vancomycin or clindamycin where MRSA is prevalent**; add cefotaxime/ceftriaxone in neonates and where Gram-negatives or Kingella are possible).
- **Short IV course with early oral switch** (current evidence, e.g. the Finnish trials of Peltola et al.): switch to high-dose oral antibiotics after about **2-4 days** once the child is clinically improving (afebrile, less pain) and **CRP is falling** (about 50% reduction).
- **Total duration: septic arthritis about 2-3 weeks**; **uncomplicated AHO about 3-4 weeks**; longer (6 weeks or more) for MRSA, complicated disease, neonates, pelvic and spinal infection, slow response.

Surgery

- **Septic arthritis of the hip is a surgical emergency: arthrotomy and thorough washout** (anterior approach; some use arthroscopy) as soon as possible. Other joints (knee, ankle, elbow) may be managed by arthroscopic washout or repeated aspiration in selected cases.
- **Osteomyelitis: operate** for a **subperiosteal or intraosseous abscess** (drill the cortex, drain the pus), **no**

clinical improvement after 48-72 hours of appropriate antibiotics, sequestrum, or concomitant septic arthritis.

- Immobilise briefly for comfort (splint); early mobilisation once pain allows. Watch for **pathological fracture** in extensive osteomyelitis (protect with a cast).

6. Complications

- **Physeal arrest** leading to limb-length discrepancy or angular deformity.
- **AVN of the femoral head** and **hip dislocation or destruction** after septic hip (the neonatal destroyed hip is **Tom Smith arthritis**; Hunka and Choi classifications of sequelae).
- **Chronic osteomyelitis**, **pathological fracture**, recurrence.
- **DVT and septic emboli** (PVL-MRSA), sepsis and multi-organ failure.
- **Joint stiffness, early arthritis**, cartilage loss.

7. Subacute and chronic osteomyelitis

- **Subacute osteomyelitis (Brodie abscess):** low virulence or partial treatment; insidious pain, few systemic signs, often normal markers.
  - Radiograph: well-defined metaphyseal (or epiphyseal) lucency with a **sclerotic rim**, sometimes crossing the physis.
  - **Differentials: osteoid osteoma, Langerhans cell histiocytosis, chondroblastoma (epiphysis), Ewing sarcoma (diaphysis).**
  - **Roberts classification** of subacute osteomyelitis (types by location and appearance); biopsy and curettage when the diagnosis is uncertain.
- **Chronic osteomyelitis: sequestrum, involucrum, cloaca, sinus**.

  - **Treatment: sequestrectomy and debridement** to healthy bleeding bone (**paprika sign**), but **wait until the involucrum is strong enough** to support the limb (otherwise fracture or segmental loss); dead-space management (antibiotic beads or cement, bone graft later), soft-tissue cover, prolonged antibiotics guided by deep cultures.
  - Long-standing sinus: watch for **Marjolin ulcer (squamous cell carcinoma)** in adults.
  - **Cierny-Mader** staging (anatomical type I-IV and host A, B, C) is used in adults.
- **Chronic recurrent multifocal osteomyelitis (CRMO; chronic non-bacterial osteomyelitis):** autoinflammatory, **sterile cultures**, recurrent multifocal bone pain (**medial clavicle**, metaphyses, spine), associated **palmoplantar pustulosis, psoriasis or inflammatory bowel disease** (SAPHO in adults). Whole-body MRI; **NSAIDs** first line, then bisphosphonates or anti-TNF agents; not long-term antibiotics.

8. Special forms

- **Neonatal osteoarticular infection:** NICU babies with umbilical or central lines; multifocal; minimal signs; **combined osteomyelitis and septic arthritis** (transphyseal vessels); high rates of growth disturbance and hip destruction.
- **Discitis (toddlers, about 1-5 years):** the **disc is vascular in young children**. Refusal to walk or sit, back pain, abdominal pain. Mild fever, raised ESR/CRP. **MRI** shows disc space narrowing with endplate oedema. Treat with **antibiotics (anti-staphylococcal) and rest/brace**; biopsy rarely needed.
- **Pelvic osteomyelitis and sacroiliitis:** buttock, groin or abdominal pain, mimics septic hip or appendicitis; FABER test; MRI.

- **Puncture wound osteochondritis of the foot (Pseudomonas):** usually osteochondral (metatarsal heads, calcaneus); needs surgical debridement plus anti-pseudomonal antibiotics (ciprofloxacin, ceftazidime).
- **Sickle cell crisis vs osteomyelitis:** both cause bone pain, fever and raised markers. Infarction is far commoner; osteomyelitis is suggested by persistent fever, a single site, positive blood cultures, and fluid collections on ultrasound/MRI. **Salmonella cover** (e.g. ceftriaxone or ciprofloxacin) in empirical treatment. **Dactylitis (hand-foot syndrome)** in infants is infarction.
- **Gonococcal arthritis** (adolescents): migratory polyarthralgia, tenosynovitis, pustular skin lesions; ceftriaxone.
- **Lyme arthritis** (endemic regions): large-joint effusion with moderate pain and less fever; serology.
- **Osteoarticular tuberculosis in children (India):**
  - Insidious monoarthritis (hip, knee), **"spina ventosa"** (tuberculous dactylitis: expanded phalanx or metacarpal), spinal TB (see Spine notes).
  - **Phemister triad:** juxta-articular osteopenia, peripheral erosions, gradual joint space narrowing.
  - Diagnosis: histology (caseating granulomas), **GeneXpert/CBNAAT** (also detects rifampicin resistance), culture.
  - **Multidrug therapy** for about 12 months under the national programme; surgery for abscesses, debridement, deformity.

9. Differential diagnosis of the irritable hip in a child

Condition	Key features
Transient synovitis	Commonest cause of acute hip pain (3-8 years), often after a viral illness, mild or no fever, able to bear some weight, near-normal markers; resolves in 1-2 weeks with rest and NSAIDs
Septic arthritis	Toxic, febrile, refuses weight bearing, raised markers (Kocher, CRP)
Perthes disease	4-8 years, insidious limp, mild pain, normal markers, radiograph or MRI changes
SCFE	Adolescent, often obese, Drehmann sign
Osteomyelitis (proximal femur, pelvis)	Focal tenderness, MRI
Juvenile idiopathic arthritis	Morning stiffness, more chronic, other joints, uveitis risk
Malignancy (leukaemia, Ewing)	Night pain, systemic features, abnormal blood count, bone lesion

Exam pearl: **Metaphysis = commonest site; S. aureus = commonest organism at all ages; Kingella = 6 months-4 years (PCR, blood-culture bottles); Salmonella = sickle cell; Pseudomonas = sneaker puncture.**

Exam pearl: **MRI is the investigation of choice for osteomyelitis; radiographs lag 10-14 days. CRP is the best marker to monitor response. Septic hip = emergency arthrotomy.**